

Lecture 6

Spearman's Rank Correlation Test

Quick recap of lecture 5

1. **Friedman test** is used for **more than two matched or blocked samples** of either ordinal or non-normal quantitative data.
2. Tests the **null hypothesis** that the **medians of the k populations are the same**.
3. It achieves this by **ranking values within blocks** and calculating the **sum of ranks for each treatment**.
4. For samples with either **k or b greater than or equal to 5**, test statistic is **chi-squared** with $k-1$ degrees of freedom.

Tests of association between populations

Recall: If an association exists between two variables, no matter how the association is measured, this association cannot and should not be interpreted as implying a cause-and-effect relationship between the two variables.

In general, two variables may have an association because:

- They are interacting with each other (i.e. one or both of the variables affects the other)
- Mere coincidence
- Both variables are affected by other variables that have not been measured in the study

Learning outcomes for this lecture

- 1) Know the conditions (data type, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Spearman's Rank Correlation Coefficient test
- 2) Know the data assumptions of Spearman's test
- 3) Be able to conduct Spearman's test

Spearman Rank Correlation Coefficient Test

- When is it appropriate? When you want to measure the association between two samples/variables of ordinal or quantitative data.

- The test is an equivalent of Pearson's Correlation Coefficient Test.

- Data and Assumptions:

- Both variables are at least ordinal (possibly quantitative), and at least one variable is not normal
 - There are a total of n randomly selected paired observations

- Note: Spearman's rank correlation coefficient is interpreted the same way as Pearson's correlation coefficient, that is:

very strong negative relationship	moderate negative relationship	no linear relationship		moderate positive relationship	very strong positive relationship	
-1	-0.5	weak negative relationship	0	weak positive relationship	0.5	+1

Spearman Rank Correlation Coefficient Test

- Hypotheses:
 - $H_0: \rho_s = 0$ (no association exists between the two variables in the underlying population)
- Either $H_1: \rho_s \neq 0$ (there is an association between the two variables in the underlying population)
 - OR $H_1: \rho_s > 0$ (the correlation between the two variables is positive)
 - OR $H_1: \rho_s < 0$ (the correlation between the two variables is negative)

Spearman Rank Correlation Coefficient Test

- Calculating the Test statistic:
 - 1) Rank the populations (call them X and Y) separately
 - 2) Calculate the difference, d, within each pair of ranks: $d = \text{rank}(x_i) - \text{rank}(y_i)$

• Test Statistic:
$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$
 where n = number of pairs in the data

For large samples ($n \geq 10$), the sampling distribution of r_s is approximately normal.

The test statistic is: $z = \frac{r_s - \mu_{r_s}}{\sigma_{r_s}}$ where $\mu_{r_s} = 0$ and $\sigma_{r_s} = \sqrt{\frac{1}{n-1}} = \frac{1}{\sqrt{n-1}}$

$$\Rightarrow z = \frac{r_s - 0}{\frac{1}{\sqrt{n-1}}} = r_s \sqrt{n-1}$$

Spearman Rank Correlation Coefficient Test

Critical Region:

- Reject H_0 if $|z| \geq z_{\alpha/2}$ (two-sided test) OR
- if $z > z_{\alpha}$ (one-sided $>$ test) OR
- if $z < -z_{\alpha}$ (one-sided $<$ test)
- where z_{α} is the critical value (critical value approach) OR if the calculated p-value $\leq \alpha$ (p-value approach)

Spearman Rank Correlation Coefficient Test --- Example 1

After several semesters without much success, Pat Statstud (a student in the lowest quarter of a statistics course) decided to try and improve his performance. Pat needed to know the secret of success for university students.

After many hours of discussion with other more successful students, Pat postulated a rather radical theory: The longer one studied, the better one's grade.

To test the theory, Pat took a random sample of 35 students in an economics course and asked each to report the average amount of time he or she studied economics, and the final mark (out of 100) obtained (see results on next slide).

Test to determine whether grade and study time are positively related.

Time	Rank_Time	Mark	Rank_Mark
30	17	71	9
5	4	30	4
36	30.5	82	17.5
37	32	98	34
32	22.5	78	14
23	7	73	10.5
34	28	82	17.5
2	2.5	25	3
34	28	94	32
43	35	99	35
34	28	85	22
32	22.5	74	12
30	17	79	15
36	30.5	82	17.5
40	34	88	26
24	8.5	55	5
0	1	7	1
25	10.5	62	6

Time	Rank_Time	Mark	Rank_Mark
29	13.5	91	29.5
21	5	66	8
31	20.5	86	23
30	17	73	10.5
33	25	90	28
30	17	88	26
33	25	91	29.5
22	6	64	7
29	13.5	83	20
24	8.5	87	24
30	17	96	33
2	2.5	16	2
31	20.5	84	21
33	25	92	31
25	10.5	82	17.5
38	33	88	26
26	12	75	13

Spearman Rank Correlation Coefficient Test --- Example 1

We are wanting to test for an association between two variables of quantitative data use Spearman $\Rightarrow$ Test

H_0 : No rank correlation/association exists between the length of study time and grade

i.e. $\rho_s = 0$

H_1 : Length of study time and grade are positively rank correlated/associated

i.e. $\rho_s > 0$

$n = 35 > 10$

$\alpha = 0.05$

Rank_Time	Rank_Mark	d_i	d_i^2
17	9	8	64
4	4	0	0
30.5	17.5	13	169
32	34	-2	4
22.5	14	8.5	72.25
7	10.5	-3.5	12.25
28	17.5	10.5	110.25
2.5	3	-0.5	2.5
28	32	-4	16
35	35	0	0
28	22	6	36
22.5	12	10.5	110.25
17	15	2	4
30.5	17.5	13	169
34	26	8	64
8.5	5	3.5	12.25
1	1	0	0
10.5	6	4.5	20.25

Rank_Time	Rank_Mark	d_i	d_i^2
13.5	29.5		
5	8		
20.5	23		
17	10.5		
25	28		
17	26		
25	29.5		
6	7		
13.5	20		
8.5	24		
17	33		
2.5	2		
20.5	21		
25	31		
10.5	17.5		
33	26		
12	13		

Rank_Time	Rank_Mark	d_i	d_i^2
17	9	8	64
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28	17.5	10.5	110.25
2.5	3	-0.5	0.25
28	32	-4	16
35	35	0	0
28	22	6	36
22.5	12	10.5	110.25
17	15	2	4
30.5	17.5	13	169
34	26	8	64
8.5	5	3.5	12.25
1	1	0	0
10.5	6	4.5	20.25

Rank_Time	Rank_Mark	d_i	d_i^2
13.5	29.5	-16	256
5	8	-3	9
20.5	23	-2.5	6.25
17	10.5	6.5	42.25
25	28	-3	9
17	26	-9	81
25	29.5	-4.5	20.25
6	7	-1	1
13.5	20	-6.5	42.25
8.5	24	-15.5	240.25
17	33	-16	256
2.5	2	0.5	0.25
20.5	21	-0.5	0.25
25	31	-6	36
10.5	17.5	-7	49
33	26	7	49
12	13	-1	1

Spearman Rank Correlation Coefficient Test --- Example 1

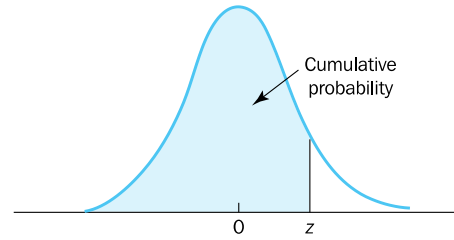
•Test Statistic Calculations: $\sum_{i=1}^n d_i^2 = 1962.5$

$$\begin{aligned} r_s &= 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} = 1 - \frac{6(1962.5)}{35(35^2 - 1)} \\ &= 1 - \frac{11775}{42840} \\ &= 1 - 0.2749 \\ &= 0.7251 \end{aligned}$$

•Test Statistic: $z = r_s \sqrt{n - 1} = (0.7251) \sqrt{34} = 4.228$

•Critical Region: Reject H_0 if $|z| \geq z_{0.05} = 1.645$, or calculated p-value ≤ 0.05

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 4.228

p-value < $1 - 0.9990$

= 0.0001

Spearman Rank Correlation Coefficient Test --- Example 1

- Conclusion (a): Since $4.228 > 1.645$ we Reject H_0 and conclude that length of study time and grade are positively correlated at the 5% significance level
- Conclusion (b): Since $p\text{-value} < 0.0001 (< 0.05)$ we Reject H_0 and conclude that length of study time and grade are positively correlated
- Q: What would we have concluded at the 2.5% significance level?

Spearman Rank Correlation Coefficient Test --- Example 1

- Conclusion (a): Since $4.228 > 1.645$ we Reject H_0 and conclude that length of study time and grade are positively correlated at the 5% significance level
- Conclusion (b): Since $p\text{-value} < 0.0001 < 0.05$ we Reject H_0 and conclude that length of study time and grade are positively correlated at the 5% significance level
- Q: What would we have concluded at the 2.5% significance level?
- Q: What would we have concluded at the 1% significance level?

Advantages and Disadvantages of Non- Parametric Tests

Advantages of Non-Parametric tests

- The tests can be used when parametric methods are inapplicable, or the validity of their assumptions is uncertain
- The tests are useful when sample sizes are small
- The assumptions are usually few and easily met
- They are not just restricted to quantitative data

Disadvantages of Non-Parametric tests

- Information is lost by ranking or taking signs. Because information is lost, nonparametric procedures tend to be less efficient or “powerful” than the equivalent parametric test (when one is appropriate for the data), i.e. for similar sample sizes and significant levels, confidence levels would be wider, and the null hypothesis may not be rejected as often.
- The theory is more complicated (and in this course we do not involve ourselves in the theory behind the tests).